

PROJECT PLANNING PHASE

Project Planning Template

Team ID	SWTID-2026-3342
Date	30.09.2026
Project Name	AI FAQ Assistant API

1. Product Backlog, Sprint Schedule and Estimation

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Project Setup	USN-1	As a developer, I can set up the Node.js and Express.js project environment.	2	High	Team Member(s)
Sprint-1	Project Setup	USN-2	As a developer, I can configure the required dependencies and project structure.	2	High	Team Member(s)
Sprint-1	Database	USN-3	As a developer, I can connect the application to MongoDB.	3	High	Team Member(s)
Sprint-1	User Management	USN-4	As a user, I can register an account using my required details.	3	High	Team Member(s)
Sprint-2	Authentication	USN-5	As a registered user, I can log in securely using my credentials.	2	High	Team Member(s)
Sprint-2	FAQ Management	USN-6	As an administrator, I can create a new FAQ.	3	High	Team Member(s)
Sprint-2	FAQ Management	USN-7	As an administrator, I can update existing FAQ information.	2	High	Team Member(s)
Sprint-2	FAQ Management	USN-8	As an administrator, I can delete FAQ information when required.	2	Medium	Team Member(s)
Sprint-3	FAQ Retrieval	USN-9	As a user, I can retrieve available FAQs through the API.	3	High	Team Member(s)
Sprint-3	FAQ Search	USN-10	As a user, I can search for relevant FAQ information.	3	High	Team Member(s)
Sprint-3	AI Assistant	USN-11	As a user, I can ask a question and receive an AI-assisted answer.	5	High	Team Member(s)
Sprint-3	AI Assistant	USN-12	As a user, I can receive relevant responses based on available FAQ information.	3	High	Team Member(s)
Sprint-4	API Testing	USN-13	As a developer, I can test all API endpoints and verify their responses.	3	High	Team Member(s)
Sprint-4	Error Handling	USN-14	As a developer, I can handle invalid requests and API errors appropriately.	2	Medium	Team Member(s)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-4	Documentation	USN-15	As a developer, I can document the API endpoints and project functionality.	2	Medium	Team Member(s)
Sprint-4	Final Testing	USN-16	As a developer, I can perform final testing and fix identified issues.	3	High	Team Member(s)

2. Sprint Planning

Sprint-1: Project Setup and Database

Main Goal: Set up the backend project and database.

Tasks

- Create Node.js project.
- Configure Express.js.
- Install required dependencies.
- Create project folder structure.
- Configure MongoDB.
- Create user model.
- Implement basic user registration.

Total Story Points: 10

Sprint-2: Authentication and FAQ Management

Main Goal: Implement authentication and FAQ management features.

Tasks

- Implement user login.
- Create FAQ model.
- Create FAQ API.
- Add new FAQ.
- Update FAQ.
- Delete FAQ.
- Validate user and FAQ data.

Total Story Points: 9

Sprint-3: FAQ Retrieval and AI Assistant

Main Goal: Implement FAQ retrieval, search and AI-assisted responses.

Tasks

- Retrieve FAQs.
- Implement FAQ search.
- Connect AI functionality.
- Process user questions.
- Generate AI-assisted answers.
- Return relevant responses through API.

Total Story Points: 14

Sprint-4: Testing, Error Handling and Documentation

Main Goal: Test and finalize the complete application.

Tasks

- Test API endpoints.
- Test authentication.
- Test FAQ CRUD operations.
- Test AI responses.
- Implement error handling.
- Fix bugs.
- Prepare API documentation.
- Perform final project testing.

Total Story Points: 10

3. Project Sprint Tracker

Sprint	Sprint Total Story Points	Duration	Sprint Start Date	Sprint End Date	Story Points Completed	Sprint Release Date
Sprint-1	10	5 Days	01.10.2026	05.10.2026	10	05.10.2026
Sprint-2	9	5 Days	06.10.2026	10.10.2026	9	10.10.2026
Sprint-3	14	5 Days	11.10.2026	15.10.2026	14	15.10.2026
Sprint-4	10	5 Days	16.10.2026	20.10.2026	10	20.10.2026

4. Velocity

The total planned work for the project is:

$10 + 9 + 14 + 10 = 43$ Story Points

The planned average velocity is:

$43 \div 4 = 10.75$ Story Points per Sprint

Therefore, the expected average team velocity is approximately **11 story points per sprint**.

5. Burndown Chart Data

The burndown chart represents the amount of work remaining during each sprint. The template describes a burndown chart as work remaining versus time.

Sprint	Planned Story Points	Remaining After Sprint
Sprint-1	43	33
Sprint-2	33	24
Sprint-3	24	10
Sprint-4	10	0